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$$\begin{aligned} & \int_{-1}^0 \frac{1}{\sqrt{1+x}} dx \\ &= \lim_{a \rightarrow -1+} \int_a^0 \frac{1}{\sqrt{1+x}} dx \\ &= \lim_{a \rightarrow -1+} \int_a^0 (1+x)^{-\frac{1}{2}} dx \end{aligned}$$

Stel $u = 1 + x \Rightarrow du = dx$

$$x = a \Rightarrow u = 1 + a$$

$$x = 0 \Rightarrow u = 1$$

$$\begin{aligned} &= \lim_{a \rightarrow -1+} \int_{1+a}^1 u^{-\frac{1}{2}} du \\ &= \lim_{a \rightarrow -1+} [2\sqrt{u}]_{1+a}^1 \\ &= \lim_{a \rightarrow -1+} (2\sqrt{1} - 2\sqrt{1+a}) \\ &= \lim_{a \rightarrow -1+} (2 - 2\sqrt{1+a}) \\ &= 2 \end{aligned}$$

De oneigenlijke integraal is convergent.

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References